

CyLab hashcrack Writeup

Challenge Overview

Event: PicoCTF 2025

Challenge Name: hashcrack

Category: Cryptography

Difficulty: Easy

Points: 100

Challenge Description

A company stored a secret message on a server which got breached due to the admin using weakly hashed passwords. Can you gain access to the secret stored within the server?

Hint 1: Understanding hashes is very crucial. [Read more here](#).

Hint 2: Can you identify the hash algorithm? Look carefully at the length and structure of each hash identified.

Hint 3: Tried using any hash cracking tools?

Note: for this challenge I'll be showing you two ways in which you can solve this challenge. The first way is going to be an in-depth approach for beginners and the other way will be explained in a few sentences for seasoned individuals.

Step 1: Accessing the Server

First, Access the server using `nc verbal-sleep.picoctf.net 62739`. Once you access the server will be greeted with the following. As you can see our hash is `482c811da5d5b4bc6d497ffa98491e38`.



```
$ nc verbal-sleep.picoctf.net 62739
Welcome!! Looking For the Secret?

We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the password for identified hash: 
```

Step 2: Identify the Hash Type Used

There are numerous ways in which you can discover what hash you are working with. I will show you two ways you can discover the hash you are trying to identify. The first way to identify a hash is to use `hash-identifier` in a terminal. The syntax for `hash-identifier` is `hash-identifier 482c811da5d5b4bc6d497ffa98491e38`.


```
$ nth --text '482c811da5d5b4bc6d497ffa98491e38'
Name-That-Hash
https://twitter.com/bee_sec_san
https://github.com/HashPals/Name-That-Hash
482c811da5d5b4bc6d497ffa98491e38
Most Likely
MD5, HC: 0 JtR: raw-md5 Summary: Used for Linux Shadow files.
MD4, HC: 900 JtR: raw-md4
NTLM, HC: 1000 JtR: nt Summary: Often used in Windows Active Directory.
Domain Cached Credentials, HC: 1100 JtR: mscach
Least Likely
Domain Cached Credentials 2, HC: 2100 JtR: mscach2 Double MD5, HC: 2600 Tiger-128, Skein-256(128),
Skein-512(128), Lotus Notes/Domino 5, HC: 8600 JtR: lotus5 md5(md5(md5($pass))), HC: 3500 Summary: Hashcat mode i
only supported in hashcat-legacy. md5(uppercase(md5($pass))), HC: 4300 md5(sha1($pass)), HC: 4400
md5(utf16($pass)), JtR: dynamic_29 md4(utf16($pass)), JtR: dynamic_33 md5(md4($pass)), JtR: dynamic_34 Haval-128,
JtR: haval-128-4 RIPEMD-128, JtR: ripemd-128 MD2, JtR: md2 Snefru-128, JtR: snefru-128 DNSSEC(NSEC3), HC: 8300
Admin v2.x, HC: 9900 JtR: radmin Cisco Type 7, BigCrypt, JtR: bigcrypt
(solomondelta@solomondelta)-[~]
$
```

Name-That-Hash also believes that MD5 is the hash being utilized in this challenge. Note, that MD4 and NTLM are also suspected hashes by Name-That-Hash. Since MD5 has appeared twice and at the top of the list for both tools let's work with the MD5 hash and see if that works.

Step 3: Create a File for the Hash

Now we want to create a file for the hash so we can crack it. To do this enter this command `echo '482c811da5d5b4bc6d497ffa98491e38' > hash.txt` in a terminal. This command takes the hash and puts it into a text document for us to crack.

```
$ echo '482c811da5d5b4bc6d497ffa98491e38' > hash.txt
```

Step 4: Utilizing Hashcat

Hashcat is a phenomenal tool for this challenge and it's exactly what we need. If you do not have Hashcat use this command `sudo apt install hashcat` to download it.

```
$ sudo apt install hashcat
```

Once hashcat has been downloaded verify you have downloaded hashcat with the command `hashcat --version`.

```
$ hashcat --version
v7.1.2
```

We already know that we are dealing a MD5 hash so now we need to use the correct hashcat syntax to crack this hash. There are numerous hashcat syntaxs you can utilize to crack this MD5 hashes but I will only be showing you one way for this challenge. You can always go to Hashcats official website <https://hashcat.net/hashcat/> to find other ways. Now to crack this MD5 hash in Hashcat use the following syntax: hashcat -m 0 -a 0 hash.txt /usr/share/wordlists/rockyou.txt.

```
-$ hashcat -m 0 -a 0 hash.txt /usr/share/wordlists/rockyou.txt
```

hashcat: lets the terminal know that we want to access and use Hashcat.

-m 0: specifies the hash mode for MD5.

-a 0: specifies the attack mode as a dictionary attack using a wordlist.

/usr/share/wordlists/rockyou.txt: is the path to the rockyou wordlist. The rockyou.txt is a wordlist already included in Hashcat that contains approximately 14.3 million passwords.

```
* Device #01: cpu-skylake-avx512-11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz, 2763/55212MCU
```

```
Minimum password length supported by kernel: 0  
Maximum password length supported by kernel: 256
```

```
Hashes: 1 digests; 1 unique digests, 1 unique salts  
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates  
Rules: 1
```

```
Optimizers applied:
```

```
* Zero-Byte  
* Early-Skip  
* Not-Salted  
* Not-Iterated  
* Single-Hash  
* Single-Salt  
* Raw-Hash
```

```
ATTENTION! Pure (unoptimized) backend kernels selected.  
Pure kernels can crack longer passwords, but drastically reduce performance.  
If you want to switch to optimized kernels, append -O to your commandline.  
See the above message to find out about the exact limits.
```

```
Command Breakdown:
```

```
Watchdog: Temperature abort trigger set to 90c
```

```
Host memory allocated for this attack: 515 MB (4179 MB free)
```

```
Dictionary cache hit:: Specifies the attack mode as a dictionary attack (using a wordlist).
```

```
* Filename..: /usr/share/wordlists/rockyou.txt  
* Passwords.: 14344385  
* Bytes.....: 139921507  
* Keyspace..: 14344385
```

```
482c811da5d5b4bc6d497ffa98491e38
```

```
Session.....: hashcat  
Status.....: Cracked  
Hash.Mode.....: 0 (MD5)  
Hash.Target.....: 482c811da5d5b4bc6d497ffa98491e38  
Time.Started.....: Thu Jul 23 18:37:02 2026 (0 secs)  
Time.Estimated...: Thu Jul 23 18:37:02 2026 (0 secs)  
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)  
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)  
Guess.Queue.....: 1/1 (100.00%)  
Speed.#01.....: 6429.0 kH/s (0.48ms) @ Accel:1024 Loops:1 Thr:1 Vec:16  
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)  
Progress.....: 12288/14344385 (0.09%)  
Rejected.....: 0/12288 (0.00%)  
Restore.Point....: 0/14344385 (0.00%)  
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-1  
Candidate.Engine.: Device Generator  
Candidates.#01...: 123456 → hawkeye  
Hardware.Mon.#01.: Temp: 37c Util: 8%
```

```
Started: Thu Jul 23 18:37:01 2026  
Stopped: Thu Jul 23 18:37:04 2026
```

Once you have run this command in the terminal hashcat will crack the hash displaying the password as depicted in the screenshot above.

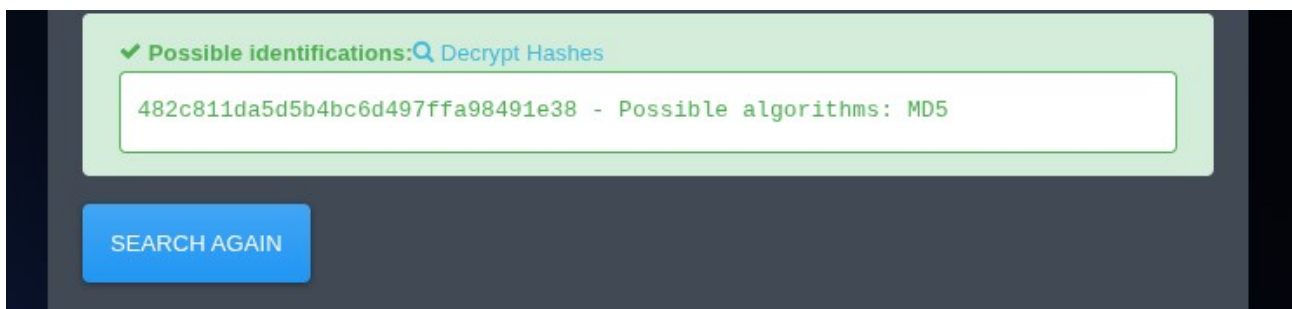
Easier Way to Solve This Challenge

As I promised you in the beginning I will now show you an easier way to solve this challenge. First go to a website like www.hashes.com. Scroll to the bottom of the website and select Hash Identifier underneath the tools section.



dgsg

Submit the hash in the display box and the results will appear.



dgg

After you do this go back to the homepage, insert the hash in the display box, solve the captcha, and then press submit and search to crack your hash.

Hashes.com

Decrypt MD5, SHA1, MySQL, NTLM, SHA256, MD5 Email, SHA256 Email, SHA512 hashes

Enter your hashes here and we will attempt to decrypt them for free online.

Hashes (max. 25 separated by newline, format 'hash[:salt]') ([QEscrow](#))

482c811da5d5b4bc6d497ffa98491e38

☐ Show plains and salts in hex format

☐ Show algorithm of founds

Captcha Check

SUBMIT & SEARCH

Once you complete these steps the cracked hash will be displayed. Note, to make this challenge easier you could have skipped the hash identifier section and just entered the hash directly on the homepage.

Hashes.com

Proceeded!

1 hashes were checked: 1 found 0 not found

Found:

482c811da5d5b4bc6d497ffa98491e38

SEARCH AGAIN

Closing Thoughts

In conclusion, the hashcrack challenge was rewarding. In this challenge you learn how to identify hashes, terminal syntax, how to use hashcat, and you learned about a website that will crack the hash instantly for you. All of these are important skills to learn as you will utilize these skills in a CTF, homelab, or in a real-world environment. I know there are other ways to solve this challenge such as CrackStation.com and john the ripper but what are other techniques you would have utilized to solve this challenge?

Securing the digital realm one computer at a time.

Best Regards,

Dylan